Library Management System

A Console-Based Java Application for Managing Books

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Why This System Was Built

Problem Statement

Traditional library record-keeping using paper logs or spreadsheets is inefficient, error-prone, and difficult to scale. Libraries struggle with:

No Tracking of Book Availability

Librarians cannot instantly know if a book is issued or available.

Manual Issue & Return Records

Paper-based systems lead to data loss and mismanagement.

Scalability Issues

Managing a large collection manually becomes unsustainable.

Project Objectives

Automate Book Management

Add, display, issue, and return books through a simple menu interface.

Apply OOP Concepts

Demonstrate encapsulation, classes, and objects in a real-world context.

Maintain Book Status

Track whether each book is **Available** or **Issued** at any time.

Provide a User-Friendly Interface

Deliver a clear, numbered console menu for all operations.

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Who Is This System For?

The Library Management System is designed to serve users across academic and small-scale library environments who need a lightweight, efficient book tracking solution.

School & College Librarians

Manage daily book issue and return operations without complex software.

Students (Borrowers)

Interact with the system to borrow and return books by entering a Book ID.

Java Learners & Developers

Study and understand the practical application of OOP concepts in Java.

System Administrators

Add new books to the library catalog and maintain the collection inventory.

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What the System Can Do

The Library Management System provides five key operations accessible through a numbered console menu, covering the complete lifecycle of a library book.

1

Add Book

Register a new book with a unique **ID**, **Title**, and **Author**. Books are stored in an `ArrayList` within the Library.

2

Show All Books

Display the complete book catalog with each book's ID, Title, Author, and current status — **Available** or **Issued**.

3

Issue Book

Mark a book as **Issued** by its ID. The system validates availability and rejects already-issued books.

4

Return Book

Mark a previously issued book as **Available** again. Validates that the book was actually issued before accepting the return.

5

Exit

Gracefully terminate the program using a menu option, returning control to the operating system.

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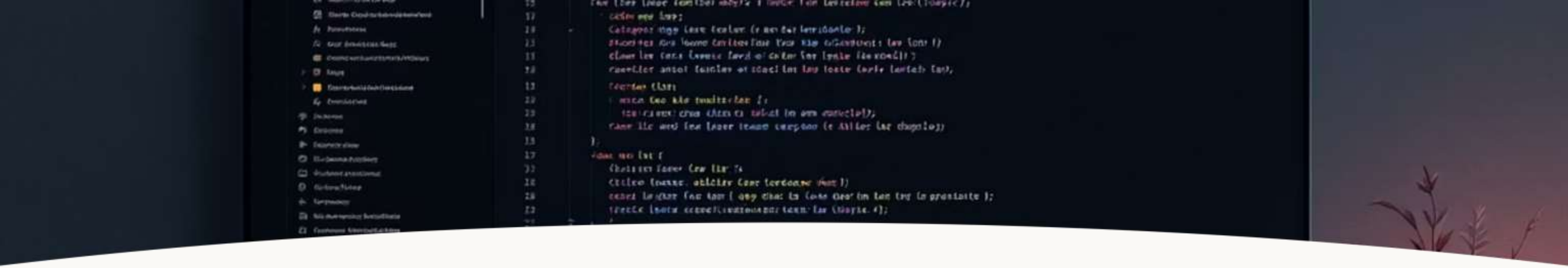

Object-Oriented Programming Concepts Used

This project is a practical demonstration of core OOP principles applied in Java to model real-world library operations.

Encapsulation

All fields in the `Book` class (`id`, `title`, `author`, `isIssued`) are declared **private**. Public getter and setter methods control access.

Classes & Objects

Two primary classes — `Book` and `Library` — model real-world entities. Each `Book` instance is a distinct object with its own state.

Constructors

The `Book` constructor initializes every new book object with `id`, `title`, `author`, and a default `isIssued = false`.

Collections (ArrayList)

The `Library` class uses an `ArrayList<Book>` to store and manage a dynamic collection of `Book` objects at runtime.

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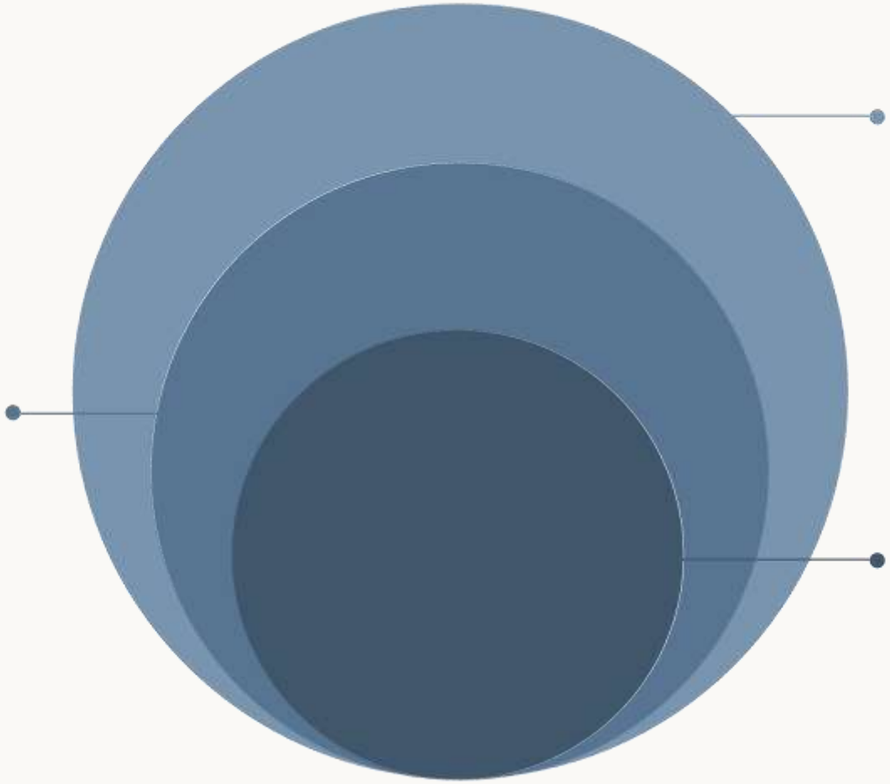
System Architecture

The system follows a simple three-layer architecture with clear separation between the data model, business logic, and user interface.



Data Model

Book class: id, title, author, isIssued

Business Logic

Library class: add/show/issue/return

User Interface

Main class, Scanner, console menu

The **Main** class acts as the entry point and UI controller, the **Library** class manages the collection and business rules, and the **Book** class represents the core data entity — each with clearly defined responsibilities.

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Code Structure Overview

The entire project is organized into three classes within a single Java file, each responsible for a distinct role in the application.

Class: `Book`

Fields

`private int id` · `private String title` · `private String author` · `private boolean isIssued`

Constructor

`Book(int id, String title, String author)` — sets all fields; `isIssued` defaults to `false`.

Methods

`getId()` · `isIssued()` · `issueBook()` · `returnBook()` · `display()`

Class: `Library`

Field

`private ArrayList<Book> books` — stores the full catalog.

Methods

`addBook(Book)` · `showBooks()` · `issueBook(int id)` · `returnBook(int id)`

Class: `Main` (Entry Point)

```
public static void main(String[] args) {
  Scanner sc = new Scanner(System.in);
  Library lib = new Library();

  while (true) {
    System.out.println(
      "1.Add 2.Show 3.Issue
       4.Return 5.Exit");
    int ch = sc.nextInt();

    switch (ch) {
      case 1: // Add new Book
      case 2: // Show all Books
      case 3: // Issue a Book
      case 4: // Return a Book
      case 5: return; // Exit
      default:
        System.out.println(
          "Invalid choice");
    }
  }
}
```

The `while(true)` loop keeps the menu active until the user explicitly selects **option 5 (Exit)**. Input is handled via `Scanner` and `sc.nextLine()` is used after `nextInt()` to consume the leftover newline character.

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System Output & Working

Below is a walkthrough of the console interaction when a user runs the Library Management System, demonstrating each of the five menu operations.

Console Menu & Sample Session

```
1.Add  2.Show  3.Issue  4.Return  5.Exit
> 1
ID: 101
Title: Java Programming
Author: James Gosling
Book added!

1.Add  2.Show  3.Issue  4.Return  5.Exit
> 1
ID: 102
Title: Data Structures
Author: Mark Allen Weiss
Book added!

1.Add  2.Show  3.Issue  4.Return  5.Exit
> 2
101 | Java Programming | James Gosling | Available
102 | Data Structures  | Mark Allen Weiss | Available

1.Add  2.Show  3.Issue  4.Return  5.Exit
> 3
Enter ID: 101
Book issued!

1.Add  2.Show  3.Issue  4.Return  5.Exit
> 2
101 | Java Programming | James Gosling | Issued
102 | Data Structures  | Mark Allen Weiss | Available

1.Add  2.Show  3.Issue  4.Return  5.Exit
> 4
Enter ID: 101
Book returned!

1.Add  2.Show  3.Issue  4.Return  5.Exit
> 5
```

Output Logic Explained

Add Book (Case 1)

Prompts for ID, Title, Author → creates a `new Book()` → appends to `ArrayList` → prints **"Book added!"**

Show Books (Case 2)

Iterates the `ArrayList` → calls `display()` on each `Book` → prints ID, Title, Author, and **Issued/Available** status.

Issue Book (Case 3)

Finds book by ID → checks `!isIssued()` → calls `issueBook()` → sets `isIssued = true` → prints **"Book issued!"** or **"Book not available."**

Return Book (Case 4)

Finds book by ID → checks `isIssued()` → calls `returnBook()` → sets `isIssued = false` → prints **"Book returned!"** or **"Invalid return."**

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Conclusion

The **Library Management System** successfully demonstrates how Object-Oriented Programming principles in Java can model and solve a real-world problem in a clean, maintainable way.

OOP in Action

The project puts encapsulation, classes, constructors, and collections to practical use — bridging academic concepts with real implementation.

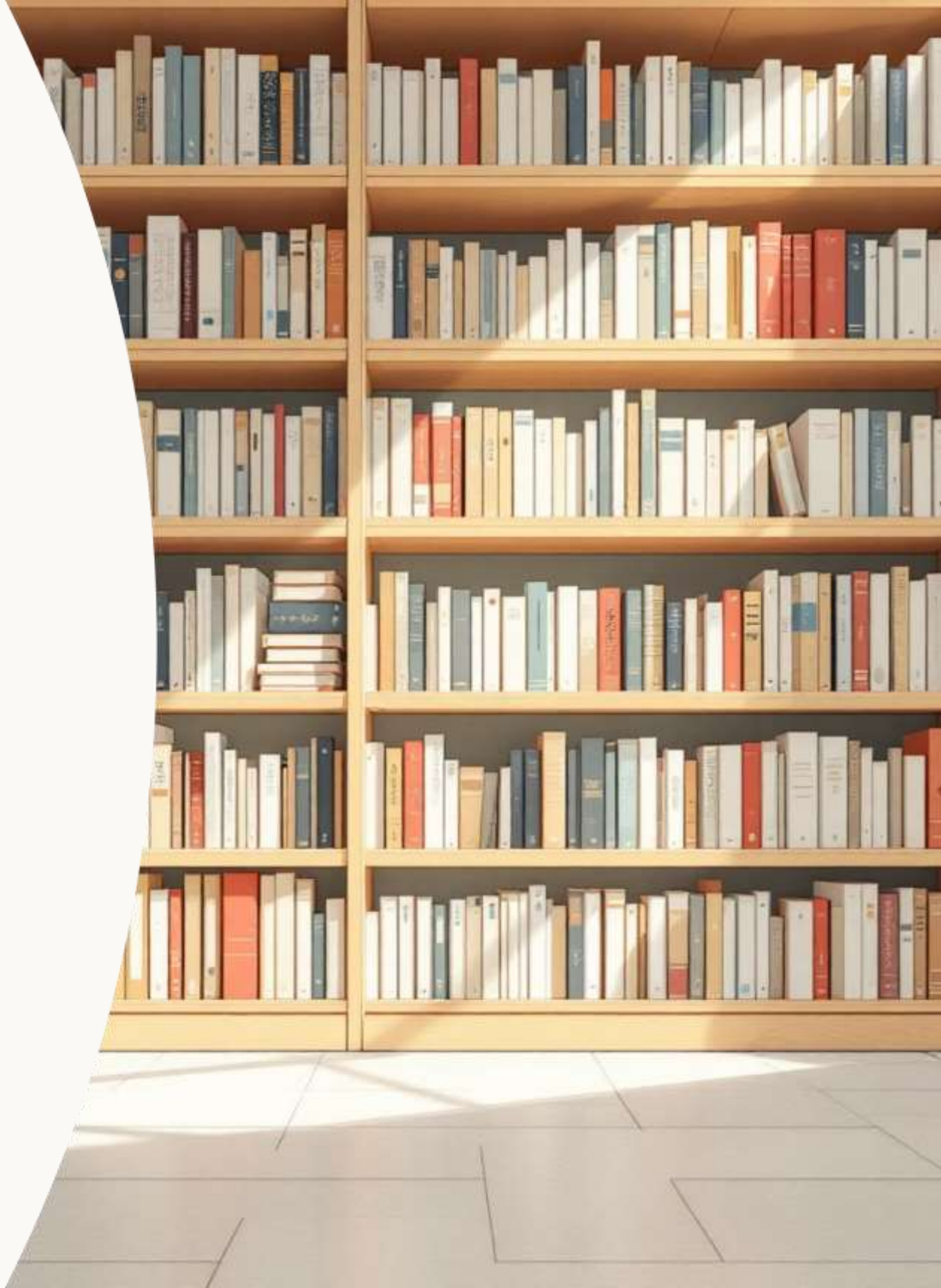
Complete Functionality

All core library operations — adding, viewing, issuing, and returning books — are handled reliably with proper input validation.

Scalable Foundation

The modular class structure makes it straightforward to extend — adding due dates, member profiles, fine calculation, or a GUI in future iterations.

This project lays a strong foundation for more advanced systems — the same OOP architecture can be extended into database-connected or web-based library platforms.



Thank You

Library Management System

We appreciate your time and attention. This project demonstrates a complete, functional Java application built on solid Object-Oriented Programming principles — from a clean data model to a fully interactive console interface.

Language

Java (JDK)

Paradigm

Object-Oriented Programming

Interface

Console / Terminal

Classes

Book · Library · Main